Indices, Surds & Quadratics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Laws of Indices & Surd Arithmetic**

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.

The points of intersection satisfy both equations. Therefore, we can set the equations equal to each other: $2x + c = -x^2 + 4x + 5$ Rearranging this equation gives: $x^2 - 2x + (c - 5) = 0$ We are given that the graphs intersect at $x = -1$. Substituting $x = -1$ into the quadratic equation $y = -x^2 + 4x + 5$: $y = -(-1)^2 + 4(-1) + 5 = -1 - 4 + 5 = 0$ So, the point of intersection is $(-1, 0)$. Since this point also lies on the line $y = 2x + c$, we can substitute $x = -1$ and $y = 0$ into the linear equation.

2 **[Foundational]** • **Target: Completing the Square & Turning Points**

The points A, B, and C have coordinates A(2, 5), B(k, 1), and C(8, -3). If A, B, and C are collinear, find the value of k.

For three points A, B, and C to be collinear, the gradient of the line segment AB must be equal to the gradient of the line segment BC (or AC). The gradient formula between two points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$. Gradient of AB, $m_{AB} = \frac{1-5}{k-2} = \frac{-4}{k-2}$. Gradient of BC, $m_{BC} = \frac{-3-1}{8-k} = \frac{-4}{8-k}$. Equating the gradients: $\frac{-4}{k-2} = \frac{-4}{8-k}$ Since the numerators are equal and non-zero, the denominators.

3 **[Foundational]** • **Target: The Discriminant & Nature of Roots**

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

The original points are A(1, 2), B(4, 2), and C(2, 5). The curve $y = f(x)$ is a parabola. Since A and B have the same y-coordinate, the axis of symmetry is $x = \frac{1+4}{2} = 2.5$. The equation of the parabola can be written as $y = a(x - 2.5)^2 + k$. Substituting C(2, 5): $5 = a(2 - 2.5)^2 + k \Rightarrow 5 = 0.25a + k$. Step 1: Reflection in the line $y = x$. This transformation swaps the x and y coordinates. A'(2, 1), B'(2, 4), C'(5, 2). Step 2: Translation by the vector $(-3; 0)$. This subtracts 3.

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4 [Foundational] • Target: Disguised Quadratics & Exam-Style Modeling

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A?

The original coordinates of point A are (-4, 2). A translation of 3 units to the right means we add 3 to the x-coordinate. A translation of 2 units down means we subtract 2 from the y-coordinate. New x-coordinate = Original x-coordinate + 3 = -4 + 3 = -1 New y-coordinate = Original y-coordinate - 2 = 2 - 2 = 0 Therefore, the new coordinates of point A are (-1, 0).

5 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Given the quadratic equation $x^2 - 5x + 6 = 0$, find the sum of the roots.

For a quadratic equation in the form $ax^2 + bx + c = 0$, the sum of the roots is given by the formula $-\frac{b}{a}$. In the given equation $x^2 - 5x + 6 = 0$, we have $a = 1$, $b = -5$, and $c = 6$. Therefore, the sum of the roots is $-\frac{-5}{1} = 5$.

6 [Foundational] • Target: Completing the Square & Turning Points

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

7 [Foundational] • Target: The Discriminant & Nature of Roots

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

Let X be the weight of a randomly selected bag of sugar. We are given that $X \sim N(\mu = 1000, \sigma^2 = 5^2)$. We want to find $P(990 < X < 1010)$. First, we convert the weights to z-scores: For $X = 990$, $z = \frac{990-1000}{5} = \frac{-10}{5} = -2$. For $X = 1010$, $z = \frac{1010-1000}{5} = \frac{10}{5} = 2$. So we need to find $P(-2 < Z < 2)$, where Z is the standard normal distribution. This probability can be found using a standard normal distribution table or calculator. The area to the left of $z = 2$ is.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Disguised Quadratics & Exam-Style Modeling**

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.

The formula for the arc length of a sector is $L = r\theta$, where L is the arc length, r is the radius, and θ is the angle in radians. We are given that $L = 6\pi$. So, we have the equation: $r\theta = 6\pi$ (1) The formula for the area of a sector is $A = \frac{1}{2}r^2\theta$, where A is the area, r is the radius, and θ is the angle in radians. We are given that $A = 27\pi$. So, we have the equation: $\frac{1}{2}r^2\theta = 27\pi$ (2) We can rewrite equation (2) as $r(r\theta) = 54\pi$. Sub.

9 **[Intermediate]** • **Target: Laws of Indices & Surd Arithmetic**

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.

To find the gradient of the tangent to the curve at $P(3, 2)$, we determine the derivative $\frac{dy}{dx}$ and evaluate it at $x = 3$. 1. Differentiate $y = x^2 - 4x + 5$ with respect to x using the power rule: $\frac{dy}{dx} = 2x - 4$ 2. Substitute $x = 3$ into the derivative: $\frac{dy}{dx}\Big|_{x=3} = 2(3) - 4 = 6 - 4 = 2$ Common errors: - 6 occurs if only the first term is differentiated (giving $2x$) and evaluated at $x = 3$, omitting the deriv.

10 **[Intermediate]** • **Target: Completing the Square & Turning Points**

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.

The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{5} P(X = x) = 1$. Given $P(X = x) = kx$, we have: $P(X = 1) = k(1) = k$ $P(X = 2) = k(2) = 2k$ $P(X = 3) = k(3) = 3k$ $P(X = 4) = k(4) = 4k$ $P(X = 5) = k(5) = 5k$ Summing these probabilities: $k + 2k + 3k + 4k + 5k = 1$ $15k = 1$ $k = \frac{1}{15}$ Therefore, the value of k is $\frac{1}{15}$.

11 **[Intermediate]** • **Target: The Discriminant & Nature of Roots**

Solve $\sin(x) = \frac{1}{2}$ for $0^\circ \le x \le 360^\circ$.

The reference angle gives one solution and CAST-rule symmetry gives the second: $x = 30^\circ$ or $x = 150^\circ$ within $[0^\circ, 360^\circ]$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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12 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.

The formula relating arc length (s), radius (r), and angle (θ) in radians is $s = r\theta$. We are given $s = 6$ cm and $r = 3$ cm. Substituting these values into the formula, we get $6 = 3\theta$. To find θ, we divide both sides by 3: $\theta = \frac{6}{3} = 2$ radians. Therefore, the angle at the centre is USD 2 radians.

13 [Intermediate] • Target: Laws of Indices & Surd Arithmetic

A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.

To find the gradient of the curve, we need to find the first derivative of the equation with respect to x. The derivative of $y = x^3 - 3x^2 + 2x$ is $\frac{dy}{dx} = 3x^2 - 6x + 2$. To find the gradient at $x = 2$, substitute $x = 2$ into the derivative: $\frac{dy}{dx}|_{x=2} = 3(2)^2 - 6(2) + 2 = 3(4) - 12 + 2 = 12 - 12 + 2 = 2$.

14 [Intermediate] • Target: Completing the Square & Turning Points

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.

The coordinates of A are (2, 1) and the coordinates of C are (5, 5). Using the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Length of AC = $\sqrt{(5-2)^2 + (5-1)^2}$ Length of AC = $\sqrt{(3)^2 + (4)^2}$ Length of AC = $\sqrt{9 + 16}$ Length of AC = $\sqrt{25}$ Length of AC = 5.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 6.01 Differentiation & Integration

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

This is a binomial distribution scenario. The number of trials is $n = 3$ (the number of coin tosses). The probability of success (getting a head) in a single trial is $p = \frac{1}{2}$, and the probability of failure (getting a tail) is $q = 1 - p = \frac{1}{2}$. We want to find the probability of getting exactly $k = 2$ heads. The binomial probability formula is $P(X = k) = \binom{n}{k} p^k q^{n-k}$. Substituting the values: $P(X = 2) = \binom{3}{2} (\frac{1}{2})^2 (\frac{1}{2})^{3-2}$.

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16 **[Challenging]** • **Target: Disguised Quadratics & Exam-Style Modeling** • **Cross-Lesson: 7.01 Statistics & Probability**

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

We are given the velocity function $v(t) = \frac{1}{2}t^2 - 3t + 5$. Since displacement s is the integral of velocity with respect to time, we have: $s(t) = \int v(t)\,dt = \int \left(\frac{1}{2}t^2 - 3t + 5\right)dt$ Integrating term by term, we get: $s(t) = \frac{1}{2} \cdot \frac{t^3}{3} - 3 \cdot \frac{t^2}{2} + 5t + K$ $s(t) = \frac{1}{6}t^3 - \frac{3}{2}t^2 + 5t + K$ Comparing this with the given form $s(t) = At^3 + Bt^2 + Ct + D$, we can identify $A = \frac{1}{6}$, $B = -\frac{3}{2}$, $C = 5$. The con.

17 **[Challenging]** • **Target: Laws of Indices & Surd Arithmetic** • **Cross-Lesson: 8.01 Functions: Composite & Inverse**

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

The binomial expansion of $(1 + x)^n$ is given by $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$. For the given expansion $(1 + kx)^n$, we can substitute $y = kx$ to get $(1 + y)^n$. The first three terms of the expansion are $1 + ny + \frac{n(n-1)}{2!}y^2 + \ldots$. Substituting $y = kx$ back, we have: $1 + n(kx) + \frac{n(n-1)}{2!}(kx)^2 + \ldots$ $1 + nkx + \frac{n(n-1)}{2!}k^2x^2 + \ldots$ We are given that the first three terms are $1 + 6x + 12x^2$. Comparing the coefficients: 1. Coefficient of x: $nk = 6$ 2. Coefficient of.

18 **[Challenging]** • **Target: Completing the Square & Turning Points** • **Cross-Lesson: 9.01 Circular Measure**

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4;\, t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

We are given the acceleration vector $\mathbf{a} = (2t - 4;\, t^2 - 9)$ and the mass of the particle $m = 8$ kg. By Newton's Second Law, the net force $\mathbf{F} = m\mathbf{a}$. Therefore, $\mathbf{F} = 8(2t - 4;\, t^2 - 9) = (16t - 32;\, 8t^2 - 72)$. The question asks for the magnitude of the net force at $t = 2$ seconds. Substituting $t = 2$ into the force vector.

19 **[Challenging]** • **Target: The Discriminant & Nature of Roots** • **Cross-Lesson: 10.01 Series & Binomial Theorem**

A sector of a circle with radius r cm has a perimeter of P cm. The area of the sector is A cm^2. Prove that $P^2 \geq 16A$.

Let the angle of the sector be θ radians. The arc length of the sector is $r\theta$. The perimeter $P = 2r + r\theta$. The area $A = \frac{1}{2}r^2\theta$. We need to prove $P^2 \geq 16A$. Substituting the expressions for P and A: $(2r + r\theta)^2 \geq 16\left(\frac{1}{2}r^2\theta\right)$ $r^2(2 + \theta)^2 \geq 8r^2\theta$ Since r^2 is positive, we can divide both sides by r^2: $(2 + \theta)^2 \geq 8\theta$ $4 + 4\theta + \theta^2 \geq 8\theta$ $\theta^2 - 4\theta + 4 \geq 0$.

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20 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 11.01 Kinematics

A particle of mass 5 kg moves under the action of a variable force $\mathbf{F}$ N. The force is given by $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ N, where x is the displacement in metres from a fixed origin O. The particle moves from $x = 2$ m to $x = 4$ m. Find the work done by the force $\mathbf{F}$ during this displacement.

The work done by a variable force $\mathbf{F}$ when a particle moves from position x_1 to x_2 is given by the integral $W = \int_{x_1}^{x_2} \mathbf{F} \cdot d\mathbf{r}$. In this case, the force is $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ and the displacement is along the x-axis, so $d\mathbf{r} = dx\mathbf{i}$. The particle moves from $x_1 = 2$ m to $x_2 = 4$ m. Therefore, the work done is: $W = \int_2^4 (3x^2 - 2x)dx$ Integrating term by term: $W = \left[\frac{3x^3}{3} - \frac{2x^2}{2}\right]_2^4$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ completely.

a. $\frac{2x^2}{3y}$ ✓

b. $\frac{4x^2}{9y}$

c. $\frac{2x^3}{3y}$

d. $\frac{8x^2}{27y}$

Apply the power $1/3$ to each term: $8^{1/3} = 2$, $(x^6)^{1/3} = x^2$, $27^{1/3} = 3$, $(y^3)^{1/3} = y$. Result is $\frac{2x^2}{3y}$.

22 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Simplify $\sqrt{48} - \sqrt{12} + \sqrt{75}$ into the form $k\sqrt{3}$.

a. $7\sqrt{3}$ ✓

b. $5\sqrt{3}$

c. $9\sqrt{3}$

d. $6\sqrt{3}$

$\sqrt{48} = 4\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$, $\sqrt{75} = 5\sqrt{3}$. Then $4\sqrt{3} - 2\sqrt{3} + 5\sqrt{3} = 7\sqrt{3}$.

23 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Rationalize the denominator of $\frac{6}{3 - \sqrt{3}}$ and write in simplest form.

a. $3 + \sqrt{3}$ ✓

b. $2 + \sqrt{3}$

c. $3 - \sqrt{3}$

d. $6 + 2\sqrt{3}$

Multiply numerator and denominator by $(3 + \sqrt{3})$: $\frac{6(3+\sqrt{3})}{9-3} = \frac{6(3+\sqrt{3})}{6} = 3 + \sqrt{3}$.

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24 [Foundational] • Target: Completing the Square & Turning Points

Express $2x^2 - 12x + 11$ in the form $a(x-h)^2 + k$.

a. $2(x-3)^2 - 7$ ✓

b. $2(x-3)^2 + 11$

c. $2(x-6)^2 - 25$

d. $2(x-3)^2 - 11$

$2(x^2-6x)+11 = 2[(x-3)^2-9]+11 = 2(x-3)^2-18+11 = 2(x-3)^2-7$.

25 [Foundational] • Target: Completing the Square & Turning Points

Find the minimum value of $f(x) = 3x^2 + 18x + 20$ and the value of x at which it occurs.

a. **Minimum is** -7 **at** $x = -3$ ✓

b. Minimum is 20 at $x = 0$

c. Minimum is -7 at $x = 3$

d. Minimum is 7 at $x = -3$

$3(x^2+6x)+20 = 3[(x+3)^2-9]+20 = 3(x+3)^2-27+20 = 3(x+3)^2-7$. Minimum is -7 when $x = -3$.

26 [Foundational] • Target: Completing the Square & Turning Points

The curve $y = -(x+4)^2 + 15$ has a stationary point. State its nature and coordinates.

a. **Maximum at** $(-4, 15)$ ✓

b. Minimum at $(-4, 15)$

c. Maximum at $(4, 15)$

d. Minimum at $(4, -15)$

Since the leading coefficient is negative, the parabola opens downward, giving a maximum at $(-4, 15)$.

27 [Foundational] • Target: The Discriminant & Nature of Roots

Find the set of values of k for which the equation $x^2 + kx + (k+3) = 0$ has two distinct real roots.

a. $k < -2$ **or** $k > 6$ ✓

b. $-2 < k < 6$

c. $k \leq -2$ or $k \geq 6$

d. $k > 6$

Distinct real roots require $\Delta > 0$: $k^2 - 4(1)(k+3) > 0 \Rightarrow k^2 - 4k - 12 > 0 \Rightarrow (k-6)(k+2) > 0$. Thus $k < -2$ or $k > 6$.

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28 [Foundational] • Target: The Discriminant & Nature of Roots

The line $y = 2x + c$ is tangent to the curve $y = x^2 - 4x + 7$. Find the exact value of c.

a. -2 ✓

b. 2

c. 7

d. -6

Set $x^2 - 4x + 7 = 2x + c \Rightarrow x^2 - 6x + (7 - c) = 0$. For tangency, $\Delta = 0$:

$36 - 4(7 - c) = 0 \Rightarrow 36 - 28 + 4c = 0 \Rightarrow 4c = -8 \Rightarrow c = -2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Discriminant & Nature of Roots

Determine the condition on p such that the quadratic $px^2 + 4x + p = 0$ has no real roots, where $p > 0$.

a. $p > 2$ ✓

b. $0 < p < 2$

c. $p > 4$

d. $p < -2$

No real roots means $\Delta < 0$: $16 - 4p^2 < 0 \Rightarrow 4p^2 > 16 \Rightarrow p^2 > 4$. Since $p > 0$, $p > 2$.

30 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

Solve the disguised quadratic equation $x^4 - 5x^2 - 36 = 0$ for all real solutions.

a. $x = \pm 3$ ✓

b. $x = \pm 2$

c. $x = \pm 3, \pm 2i$

d. $x = 9$

Let $u = x^2$: $u^2 - 5u - 36 = (u - 9)(u + 4) = 0$. Since x is real, $u = x^2 = 9 \Rightarrow x = \pm 3$ ($u = -4$ has no real solutions).

31 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

Solve the equation $2^{2x} - 10(2^x) + 16 = 0$.

a. $x = 1$ **or** $x = 3$ ✓

b. $x = 2$ or $x = 8$

c. $x = 0$ or $x = 4$

d. $x = -1$ or $x = 3$

Let $u = 2^x$: $u^2 - 10u + 16 = 0 \Rightarrow (u - 2)(u - 8) = 0$. Thus $2^x = 2 \Rightarrow x = 1$, and $2^x = 8 \Rightarrow x = 3$.

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32 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

A rectangular enclosure is formed using 80 meters of fencing against a straight stone wall. What is the maximum possible area of the enclosure in square meters?

a. $800\ \text{m}^2$ ✓

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

Let width be x. Length along wall is $80 - 2x$. Area $A(x) = x(80 - 2x) = 80x - 2x^2 = -2(x - 20)^2 + 800$. The maximum area is $800\ \text{m}^2$ at $x = 20$ m.

33 [Intermediate] • Target: Laws of Indices & Surd Arithmetic

Find the discriminant of $3x^2 - 5x - 10 = 0$ and state the number of real roots.

a. $\Delta = -145$: two distinct real roots

b. $\Delta = 145$: one repeated real root

c. $\Delta = 153$: no real roots

d. $\Delta = 145$**: two distinct real roots** ✓

$\Delta = b^2 - 4ac = (-5)^2 - 4(3)(-10) = 145$, so the equation has two distinct real roots.

34 [Intermediate] • Target: Completing the Square & Turning Points

Find the discriminant of $x^2 - 4x - 7 = 0$ and state the number of real roots.

a. $\Delta = 52$: no real roots

b. $\Delta = 44$: one repeated real root

c. $\Delta = 44$**: two distinct real roots** ✓

d. $\Delta = -44$: two distinct real roots

$\Delta = b^2 - 4ac = (-4)^2 - 4(1)(-7) = 44$, so the equation has two distinct real roots.

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35 **[Intermediate]** • **Target: The Discriminant & Nature of Roots**

Find the discriminant of $x^2 - 6x - 2 = 0$ and state the number of real roots.

a. $\Delta = -44$: two distinct real roots

b. $\Delta = 44$**: two distinct real roots ✓**

c. $\Delta = 52$: no real roots

d. $\Delta = 44$: one repeated real root

$\Delta = b^2 - 4ac = (-6)^2 - 4(1)(-2) = 44$, so the equation has two distinct real roots.

36 **[Intermediate]** • **Target: Disguised Quadratics & Exam-Style Modeling**

Find the discriminant of $x^2 - 4x - 6 = 0$ and state the number of real roots.

a. $\Delta = 40$: one repeated real root

b. $\Delta = -40$: two distinct real roots

c. $\Delta = 48$: no real roots

d. $\Delta = 40$**: two distinct real roots ✓**

$\Delta = b^2 - 4ac = (-4)^2 - 4(1)(-6) = 40$, so the equation has two distinct real roots.

37 **[Intermediate]** • **Target: Laws of Indices & Surd Arithmetic**

Find the discriminant of $4x^2 - 5x - 2 = 0$ and state the number of real roots.

a. $\Delta = -57$: two distinct real roots

b. $\Delta = 57$**: two distinct real roots ✓**

c. $\Delta = 57$: one repeated real root

d. $\Delta = 65$: no real roots

$\Delta = b^2 - 4ac = (-5)^2 - 4(4)(-2) = 57$, so the equation has two distinct real roots.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Completing the Square & Turning Points • Cross-Lesson: 9.01 Circular Measure

Find the discriminant of $5x^2 - 6x - 3 = 0$ and state the number of real roots.

a. $\Delta = 104$: no real roots

b. $\Delta = 96$**: two distinct real roots ✓**

c. $\Delta = -96$: two distinct real roots

d. $\Delta = 96$: one repeated real root

$\Delta = b^2 - 4ac = (-6)^2 - 4(5)(-3) = 96$, so the equation has two distinct real roots.

39 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 10.01 Series & Binomial Theorem

Find the discriminant of $5x^2 + 8x - 5 = 0$ and state the number of real roots.

a. $\Delta = 164$: one repeated real root

b. $\Delta = 172$: no real roots

c. $\Delta = -164$: two distinct real roots

d. $\Delta = 164$**: two distinct real roots ✓**

$\Delta = b^2 - 4ac = (8)^2 - 4(5)(-5) = 164$, so the equation has two distinct real roots.

40 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 11.01 Kinematics

Find the discriminant of $x^2 + x - 3 = 0$ and state the number of real roots.

a. $\Delta = 21$: no real roots

b. $\Delta = 13$: one repeated real root

c. $\Delta = -13$: two distinct real roots

d. $\Delta = 13$**: two distinct real roots ✓**

$\Delta = b^2 - 4ac = (1)^2 - 4(1)(-3) = 13$, so the equation has two distinct real roots.

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41 [Challenging] • Target: Laws of Indices & Surd Arithmetic • Cross-Lesson: 12.01 Forces & Equilibrium

Find the discriminant of $3x^2 - 6x + 1 = 0$ and state the number of real roots.

a. $\Delta = 32$: no real roots

b. $\Delta = -24$: two distinct real roots

c. $\Delta = 24$**: two distinct real roots ✓**

d. $\Delta = 24$: one repeated real root

$\Delta = b^2 - 4ac = (-6)^2 - 4(3)(1) = 24$, so the equation has two distinct real roots.

42 [Challenging] • Target: Completing the Square & Turning Points • Cross-Lesson: 13.01 Newton's Laws

Find the discriminant of $x^2 - 7x + 6 = 0$ and state the number of real roots.

a. $\Delta = 33$: no real roots

b. $\Delta = 25$: one repeated real root

c. $\Delta = -25$: two distinct real roots

d. $\Delta = 25$**: two distinct real roots ✓**

$\Delta = b^2 - 4ac = (-7)^2 - 4(1)(6) = 25$, so the equation has two distinct real roots.

43 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 14.01 Energy, Work & Power

Find the discriminant of $2x^2 + 7x + 6 = 0$ and state the number of real roots.

a. $\Delta = 1$**: two distinct real roots ✓**

b. $\Delta = 9$: no real roots

c. $\Delta = -1$: two distinct real roots

d. $\Delta = 1$: one repeated real root

$\Delta = b^2 - 4ac = (7)^2 - 4(2)(6) = 1$, so the equation has two distinct real roots.

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44 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 15.01 Representation of Data

Find the discriminant of $x^2 + 2x + 1 = 0$ and state the number of real roots.

a. $\Delta = 0$**: one repeated real root ✓**

b. $\Delta = 8$: no real roots

c. $\Delta = 16$: one repeated real root

d. $\Delta = 0$: two distinct real roots

$\Delta = b^2 - 4ac = (2)^2 - 4(1)(1) = 0$, so the equation has one repeated real root.

45 [Challenging] • Target: Laws of Indices & Surd Arithmetic • Cross-Lesson: 16.01 Permutations & Combinations

Find the discriminant of $5x^2 + 10x + 11 = 0$ and state the number of real roots.

a. $\Delta = -120$: two distinct real roots

b. $\Delta = 120$: no real roots

c. $\Delta = -120$**: no real roots ✓**

d. $\Delta = -112$: one repeated real root

$\Delta = b^2 - 4ac = (10)^2 - 4(5)(11) = -120$, so the equation has no real roots.

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